



Neurotoxicity of BPA, BPS, and BPB for the hippocampal cell line (HT-22): An implication for the replacement of BPA in plastics[☆]

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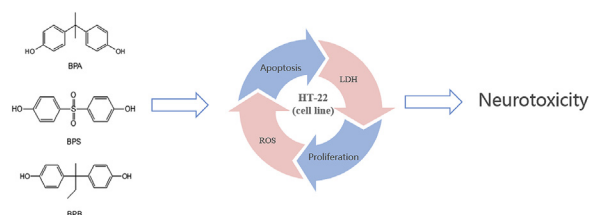
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HIGHLIGHTS

- BPA, BPS and BPB exhibit different cytotoxicity to HT-22 cells.
- Non-monotonic relationships existed between BPs levels and toxic effects of HT-22 cells.
- BPS might be surrogate to replace BPA in plastic production with its lower neurotoxicity.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 9 August 2018

Received in revised form

24 March 2019

Accepted 27 March 2019

Available online 30 March 2019

Handling Editor: A. Gies

Keywords:

Bisphenol A

Bisphenol S

Bisphenol B

Hippocampus

Neurotoxicity

ABSTRACT

Bisphenol A (BPA), a plastic additive, is ubiquitous in the environment and has endocrine disrupting effects. As many countries have prohibited the manufacture and sale of plastic products with BPA, BPA analogs have been used to replace BPA during production, including bisphenol S (BPS) and bisphenol B (BPB). To investigate the toxicities of BPA and its analogs on neurons, reactive oxygen species (ROS) assay, Annexin V-FITC (fluorescein) apoptosis detection assay, lactate dehydrogenase (LDH) cytotoxicity assay, and Cell Counting Kit-8 assay were conducted to comprehensively assess the influence of different concentrations of BPA, BPB, and BPS on ROS, apoptosis, damage, and proliferation for hippocampal HT-22 cells, respectively. Results showed that 6 h of exposure to bisphenols (BPs) could increase the ROS levels, 24 h and 48 h of exposure could induce higher apoptosis and LDH leakage rates for HT-22 cells, and 7 d of exposure could inhibit the cell proliferations. In addition, non-monotonic dose-response relationships were observed between the concentrations of bisphenols and the toxic effects mentioned above. The neurotoxic effects of BPA, BPB and BPS on HT-22 cells were in the increasing order of BPS, BPA, and BPB. In conclusion, these results showed that exposure to BPA and its analogs may result in adverse effects on hippocampal neuronal cell lines. BPS is a surrogate with lower neurotoxicity to replace BPA in production of plastic utensils.

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1. Introduction

Bisphenol A (BPA) is widely used in the production of plastic containers, canned lining, dental fillings, and thermal paper, etc. (Brunelle, 2005; Vandenberg et al., 2007; Geensa et al., 2011; Fan

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